

Assignment 1: File Descriptor Mastery

```
vboxuser@ubuntuabse:~$ ls file3.txt > stdout.txt
vboxuser@ubuntuabse:~$ cat stdout.txt
file3.txt
vboxuser@ubuntuabse:~$ cat file3.txt
hello everyone
hello all
hello error
vboxuser@ubuntuabse:~$ ls notfile.txt 2> stderr.txt
vboxuser@ubuntuabse:~$ cat stderr.txt
ls: cannot access 'notfile.txt': No such file or directory
vboxuser@ubuntuabse:~$ ls file3.txt stderr.txt &> merge.txt
vboxuser@ubuntuabse:~$ cat merge.txt
file3.txt
stderr.txt
vboxuser@ubuntuabse:~$ ls file3.txt stderr.txt > done.txt 2> error.txt
vboxuser@ubuntuabse:~$ cat done.txt
file3.txt
stderr.txt
vboxuser@ubuntuabse:~$ cat error.txt
ls: cannot access 'notfile.txt': No such file or directory
vboxuser@ubuntuabse:~$ ls file3.txt
file3.txt
vboxuser@ubuntuabse:~$ ls notfile.txt
ls: cannot access 'notfile.txt': No such file or directory
vboxuser@ubuntuabse:~$ ls stderr.txt
stderr.txt
```

Assignment 2: Pipeline Engineering

```
vboxuser@ubuntuabse:~$ touch week2/ass2/filew.txt
vboxuser@ubuntuabse:~$ cat <<EOF> week2/ass2/filew.txt
> ERROR:404 not found
> INFO:running server
> STOP: server stopped
> ERROR: 401 server error
> START: again sever started
> EOF
vboxuser@ubuntuabse:~$ cat week2/ass2/filew.txt
ERROR:404 not found
INFO:running server
STOP: server stopped
ERROR: 401 server error
START: again sever started
vboxuser@ubuntuabse:~$ grep "ERROR" week2/ass2/filew.txt | sort | uniq -c | sort -nr
      1 ERROR:404 not found
      1 ERROR: 401 server error
vboxuser@ubuntuabse:~$
```

Assignment 3: Heredoc Script Input

```
vboxuser@ubuntuabse:~$ touch script.sh
vboxuser@ubuntuabse:~$ nano script.sh
vboxuser@ubuntuabse:~$ cat <<EOF> script.sh
> #!/bin/bash
> echo "process input"
> cat <<DATA | sort | uniq -c | awk '{print "Item:", \$2, "|count:", \$1}'
> apple
> microsoft
> google
> window
> EOF
vboxuser@ubuntuabse:~$ chmod +x script.sh
vboxuser@ubuntuabse:~$ ./script.sh
process input
./script.sh: line 7: warning: here-document at line 3 delimited by end-of-file (wanted 'DATA')
Item: apple |count: 1
Item: google |count: 1
Item: microsoft |count: 1
Item: window |count: 1
vboxuser@ubuntuabse:~$
```